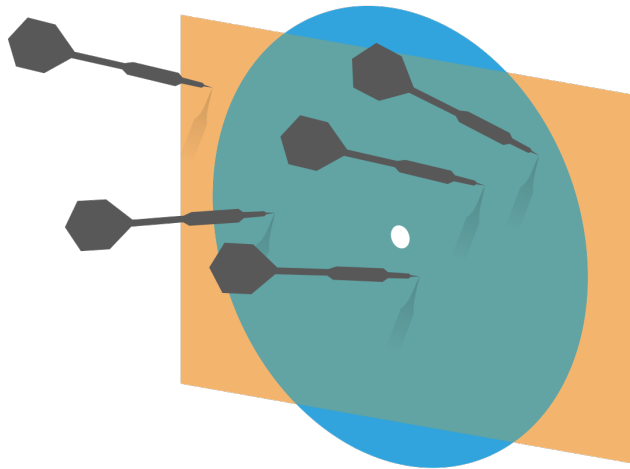


Probability and Statistics for Computer Science



Credit: wikipedia

“It’s straightforward to link a number to the outcome of an experiment. The result is a **Random variable.**” --- Prof. Forsythe

Random variable is a function, it is not the same as in **$X = X + 1$**

PollEv site



PollEv.com/hongyeliu713

Independent Vs. Disjoint

Independent : $P(A \cap B) = P(A) \cdot P(B)$

tossing two coins independently
A : 1st is H or $P(A|B) = P(A)$
B : 2nd is T or $P(B|A) = P(B)$

Disjoint : $P(A \cap B) = \emptyset$

roll a dice
ie: A is odd outcomes
B is even outcomes

Which is larger?

- ① The probability of drawing hands of 5-cards that have no pairs.
(no replacement)
from standard 52-set of cards
- ② 0.5

A ①

B. ②

$$P = \frac{|E|}{|S|} = \frac{52 \times 48 \times 44 \times 40 \times 36}{52 \times 51 \times 50 \times 49 \times 48} \approx 0.507 > 0.5$$

Last Time

✱ Conditional Probability

✱ Review

✱ Bayes rule

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

✱ Total probability

$$P(B) = \frac{P(B \cap A)}{P(B)} + P(B \cap A^c)$$

✱ Independence

$$= \sum_j P(B \cap A_j)$$

↙

$$P(B|A)P(A) + P(B|A^c) \cdot P(A^c)$$

Objectives

- * Review
- * Random Variable
- * Probability distribution
- * Cumulative distribution
- * Joint probability
- * Independence of random variables

Probability using the property of Independence: Airline overbooking (2)

- ✱ An airline has a flight with 6 seats. They always sell 8 tickets for this flight. If ticket holders show up independently with probability p , what is the probability that exactly 6 people showed up?

$$P(6 \text{ people showed up}) = \binom{8}{6} p^6 \cdot (1-p)^2$$

Probability using the property of Independence: Airline overbooking (4)

- ✱ An airline has a flight with s seats. They always sell t ($t > s$) tickets for this flight. If ticket holders show up independently with probability p , what is the probability that exactly u people showed up?

$$u \leq t$$

P(exactly u people showed up)

$$\binom{t}{u} p^u (1-p)^{t-u}$$

Probability using the property of Independence: Airline overbooking (5)

- ✱ An airline has a flight with s seats. They always sell t ($t > s$) tickets for this flight. If ticket holders show up independently with probability p , what is the probability that the flight is overbooked ?

P(overbooked)

$$\sum_{u=s+1}^t \binom{t}{u} \cdot p^u (1-p)^{t-u}$$

Independence vs Disjoint

✱ Q. Two disjoint events that have probability > 0 are certainly dependent to each other.

A. True

B. False

Independence of empty event

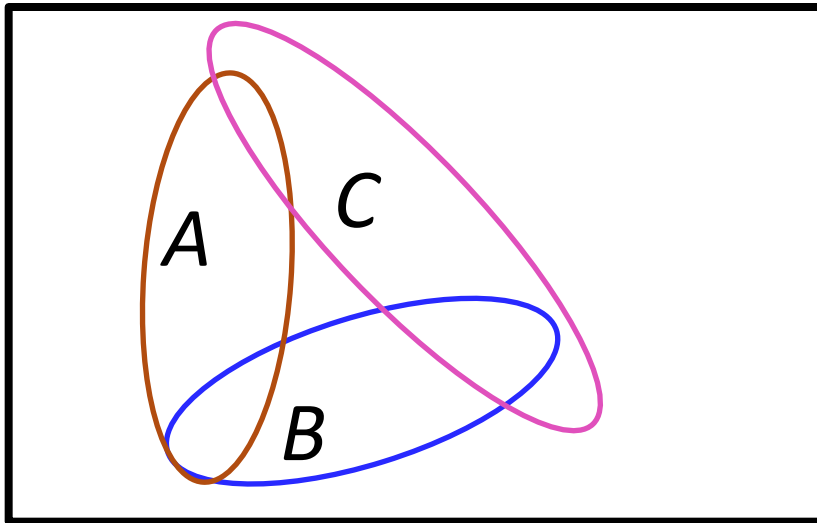
✱ Q. Any event is independent of empty event $\emptyset$.

A. True

B. False

Condition may affect Independence

- ✱ Assume event **A** and **B** are pairwise independent



Given **C**, **A** and **B** are not independent any more because they become disjoint

Conditional Independence

✱ Event **A** and **B** are conditional independent given event **C** if the following is true.

$$P(A \cap B|C) = P(A|C)P(B|C)$$

See an example in Degroot et al. Example 2.2.10

Random numbers

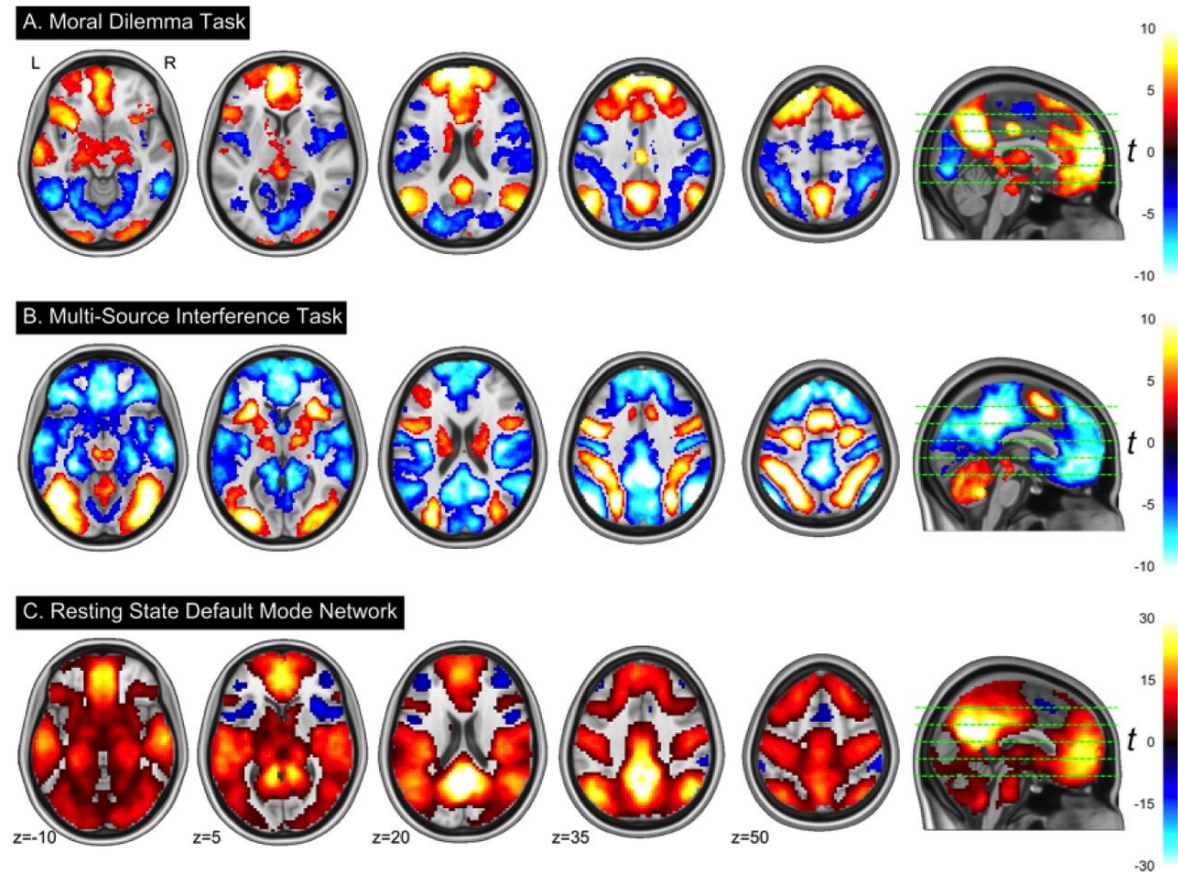
- ✱ Amount of money on a bet
- ✱ Age at retirement of a population
- ✱ Rate of vehicles passing by the toll
- ✱ Body temperature of a puppy in its pet clinic
- ✱ Level of the intensity of pain in a toothache

Random variable as vectors

Brain imaging
of Human
emotions

- A) Moral
conflict
- B) Multi-task
- C) Rest

$(x, y, z, t, \text{intensity})$



A. McDonald et al. NeuroImage doi:
10.1016/j.neuroimage.2016.10.048

Random variables

A random variable maps all outcomes to Numbers!

$$(\omega) \rightarrow (X)$$

outcomes
are disjoint

Bernoulli Random Variable



$$X(\omega) = \begin{cases} 1 & \omega = \text{Head} \\ 0 & \omega = \text{Tail} \end{cases}$$

$$\begin{aligned} P(\omega = \text{Head}) &= p & \rightarrow & P(X=1) = p \\ P(\omega = \text{Tail}) &= 1-p & & P(X=0) = 1-p \end{aligned}$$

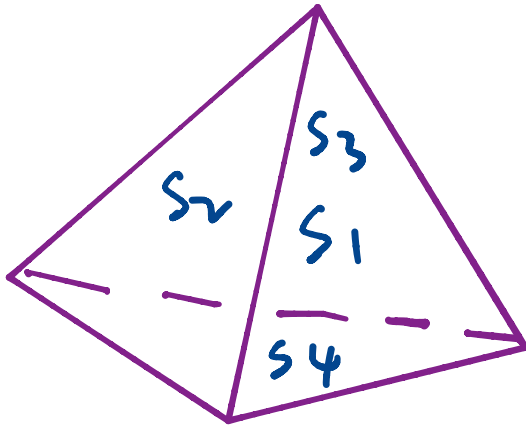
Random variables

- ✱ The values of a random variable can be either **discrete**, **continuous** or **mixed**.

Discrete Random variables

- ✱ The range of a discrete random variable is a countable set of real numbers.

4-die



$P(s_1)$

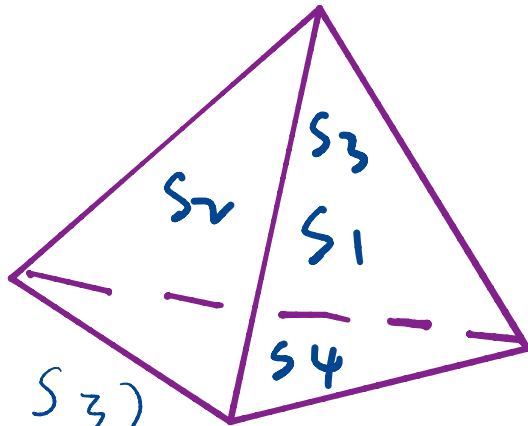
$= ?$ 0.25

ω		$X(\omega)$
s_1	$\rightarrow$	1
s_2	$\rightarrow$	2
s_3	$\rightarrow$	3
s_4	$\rightarrow$	4

Discrete Random variables

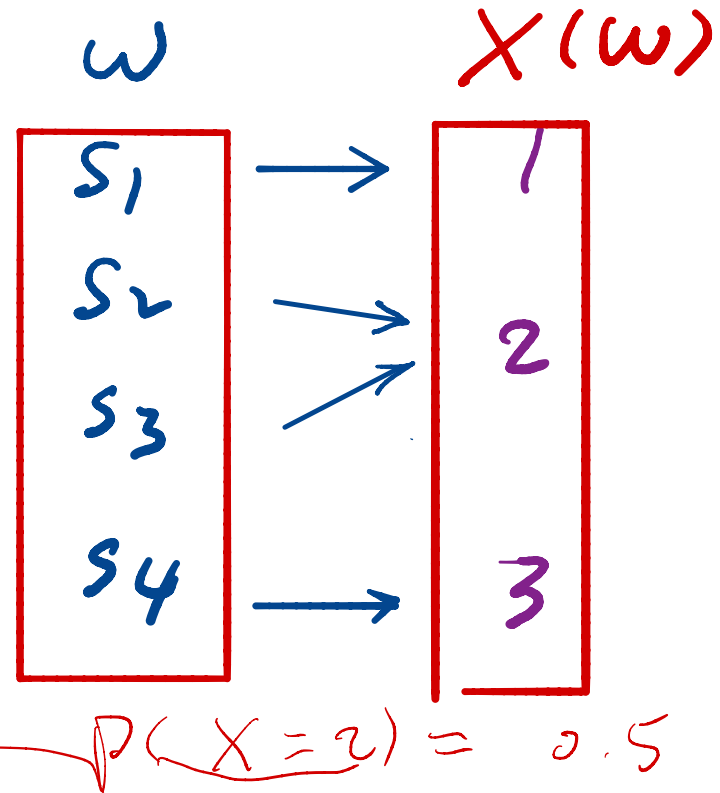
- ✱ The range of a discrete random variable is a countable set of real numbers.

4-die



$P(S_2 \text{ or } S_3)$

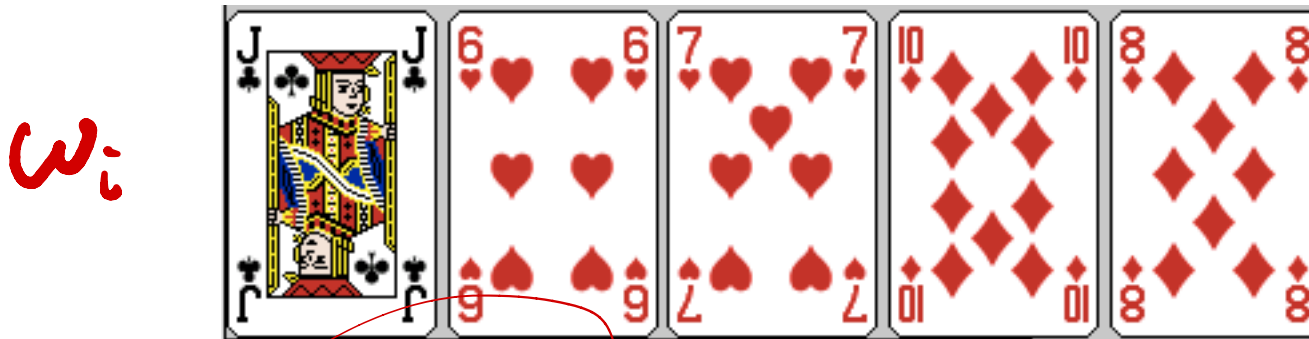
$$= P(S_2) + P(S_3) = \frac{1}{4} + \frac{1}{4} = 0.5$$



$$P(X=2) = 0.5$$

Random Variable Example

✱ Number of pairs in a hand of 5 cards



Four of
the same
→ 2 pairs

$X(\omega_i) = ?$

0

- ✱ Let a single outcome be the hand of 5 cards
- ✱ Each outcome maps to values in the set of numbers $\{0, 1, 2\}$

$P(X=0)$

$\frac{50 \times 48 \cdots}{52 \times \cdots}$

Random Variable Example

- ✱ **Number of pairs in a hand of 6 cards**
- ✱ Let a single outcome be the hand of 6 cards
- ✱ What is the range of values of this random variable?

$X(\omega)$ can take

Q: Random Variable

- ✱ If we roll a 3-sided fair die, and define random variable U , such that

$$U = \begin{cases} -1 & \omega \rightarrow \text{side 1} \\ 0 & \omega \rightarrow \text{side 2} \\ 1 & \omega \rightarrow \text{side 3} \end{cases}$$

$$X = U^2$$

$$\begin{aligned} (-1)^2 &= 1 \\ 0^2 &= 0 \\ 1^2 &= 1 \end{aligned}$$



what is the range of X ?

$$P(X=1) = \frac{2}{3}$$

A. $\{-1, 0, 1\}$

B. $\{0, 1\}$

$\rightarrow P(s_1, \text{ or } s_3)$

Three important facts of Random variables

- ✱ Random variables have **probability functions**
- ✱ Random variables can be **conditioned** on events or other random variables
- ✱ Random variables have **averages**

Random variables have probability functions

✱ Let X be a random variable

✱ The set of outcomes $\{(\omega_i \in \Omega) \text{ s.t. } X(\omega_i) = x_0\}$
is an event with probability

ω_i are
disjoint

$$P(X = x_0) = \sum_i P(\omega_i)$$

X is the discrete random variable

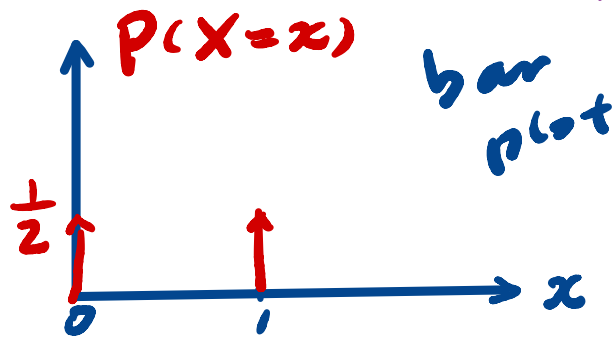
x_0 is any specific instance that X takes on

Probability Distribution

- ✱ $P(X = x)$ is called the probability distribution for all possible x
- ✱ $P(X = x)$ is also denoted as $P(x)$ or $p(x)$
- ✱ $P(X = x) \geq 0$ for all values that X can take, and is 0 everywhere else
- ✱ The sum of the probability distribution is 1
$$\sum_x P(x) = 1$$

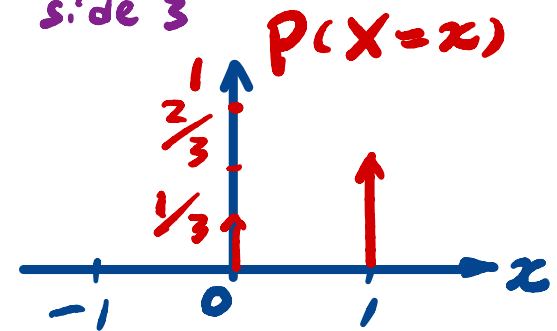
Examples of Probability Distributions

$$X(\omega) = \begin{cases} 1 & \text{head (fair coin)} \\ 0 & \text{tail} \end{cases}$$



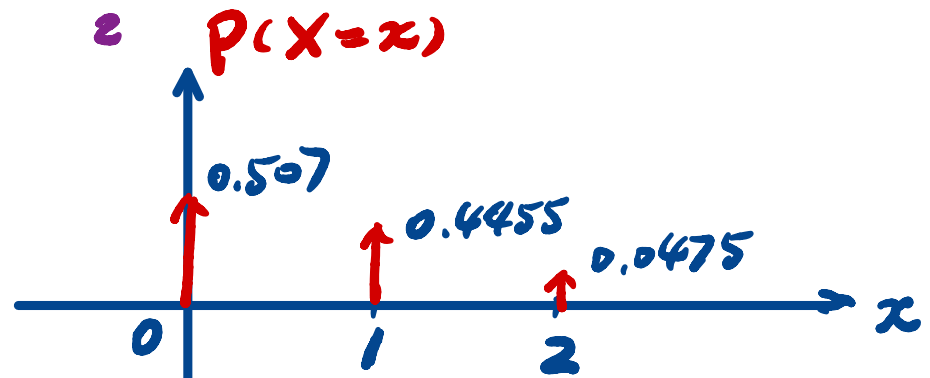
$$U(\omega) = \begin{cases} -1 & \text{side 1} \\ 0 & \text{side 2} \\ 1 & \text{side 3} \end{cases} \quad \text{fair die}$$

$$X(\omega) = U^2$$



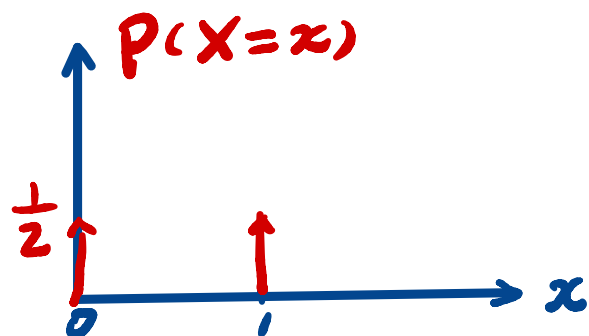
$$X(\omega) = \begin{cases} 0 & \text{\# of pairs} = 0 \\ 1 & \\ 2 & \end{cases}$$

\# pairs in a hand of 5-cards



Another way to write PDF

$$X(\omega) = \begin{cases} 1 & \text{head} \\ 0 & \text{tail} \end{cases}$$

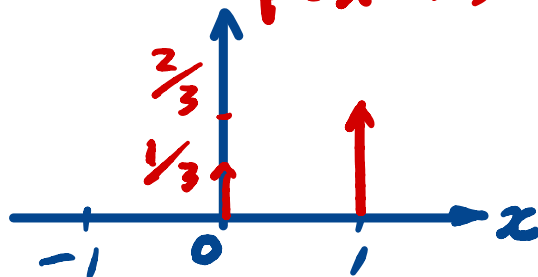


$$P(X=x) = \begin{cases} \frac{1}{2} & x=0 \\ \frac{1}{2} & x=1 \\ 0 & \text{otherwise} \end{cases}$$

$$U(\omega) = \begin{cases} -1 & \text{side 1} \\ 0 & \text{side 2} \\ 1 & \text{side 3} \end{cases}$$

$$X(\omega) = U^2$$

$P(X=x)$



$$P(X \leq 0) = \frac{2}{3}$$

$$P(X=x) = \begin{cases} \frac{1}{3} & x=0 \\ \frac{2}{3} & x=1 \\ 0 & \text{otherwise} \end{cases}$$

$P(X \leq 1) = P(X=0 \text{ or } X=1) = P(X=0) + P(X=1)$

Cumulative distribution

- ✱ $P(X \leq x)$ is called the cumulative distribution function of X
- ✱ $P(X \leq x)$ is also denoted as $f(x)$
- ✱ $P(X \leq x)$ is a non-decreasing function of x

$$\sum_{x=-\infty}^x p(x=x)$$

↓
in case

X is
discrete

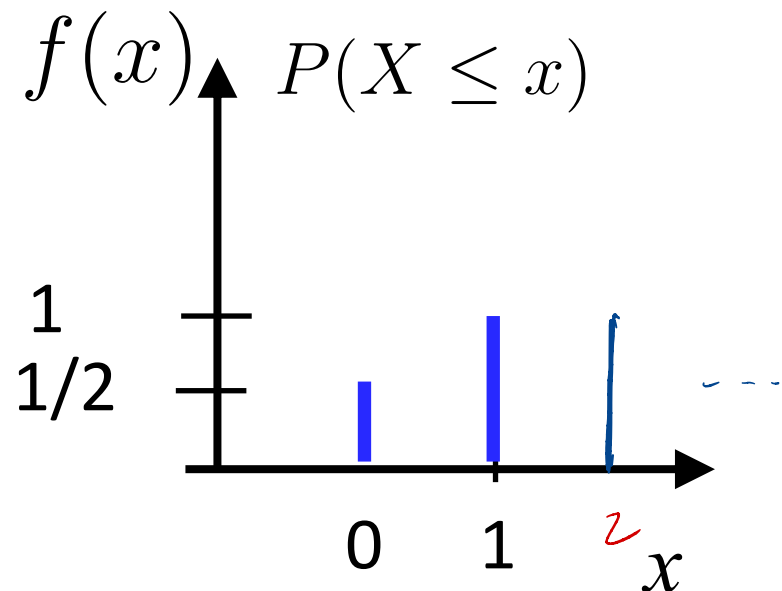
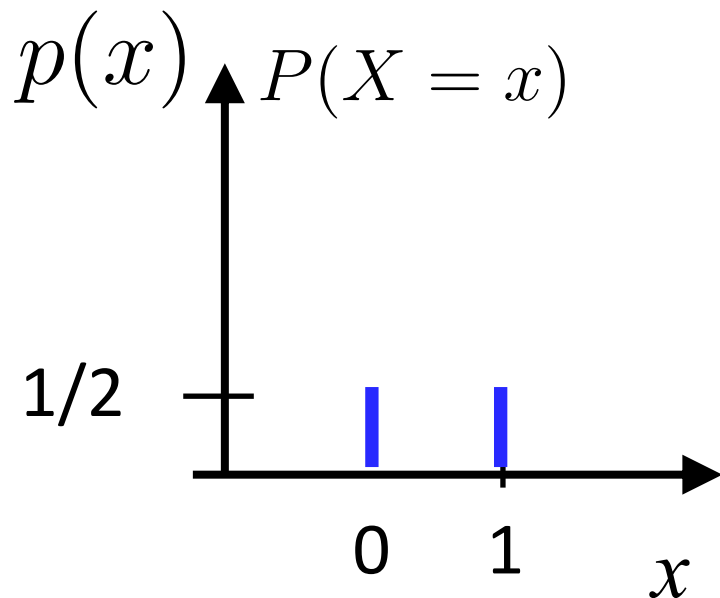
$$\lim_{x \rightarrow \infty} P(X \leq x) = 1$$

$\therefore P(\Omega) = 1$

Probability distribution and cumulative distribution

✱ Give the random variable **X** ,

$$X(\omega) = \begin{cases} 1 & \text{outcome of } \omega \text{ is head} \\ 0 & \text{outcome of } \omega \text{ is tail} \end{cases}$$



What is the value?

A biased four-sided die is rolled once.
Random variable X is defined to be the down-face value.

$$P(X=x) = \begin{cases} \frac{x}{10} & x=1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

$$P(X \leq 4)$$

A) 0.1

B) 0.3

C) 0.2

D) 0.6

$$\begin{aligned} P(X \leq 4) &= P(X=1) + P(X=2) \\ &\quad + P(X=3) + P(X=4) \\ \text{(E) } &1 \\ &\text{or } 1 - P(X > 4) \end{aligned}$$

Functions of Random Variables

$$U$$
$$X = U^2$$

$$X = |U|$$

$$S = X_1 + X_2$$

$$D = X_1 - X_2$$

Q. Do they have the same pdf?

$$X(\omega) = \begin{cases} 1 & \text{Head} \\ 0 & \text{Tail} \end{cases}$$

$$Y(\omega) = \begin{cases} 1 & \text{Head} \\ 0 & \text{Tail} \end{cases}$$

same fair coin

$$U = 2X$$

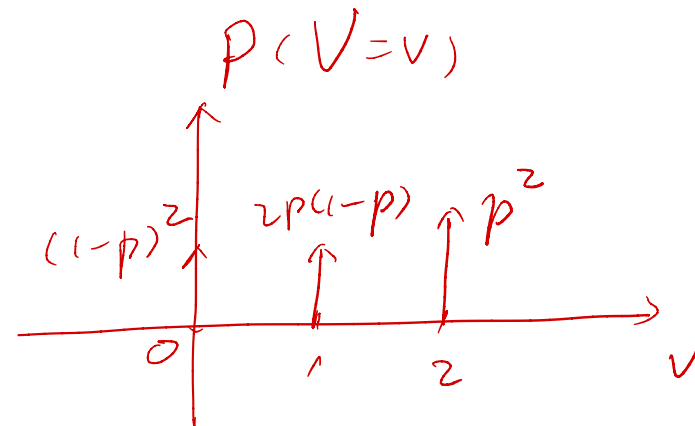
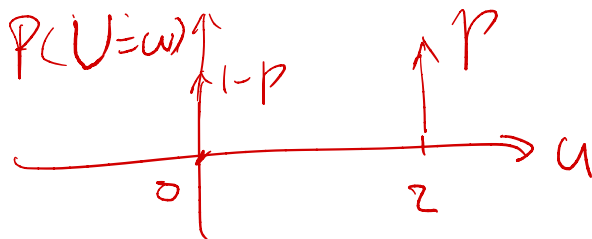
$$V = X + Y$$

Indpt
tosses

Do U and V have the same Pdf?

A) Yes

B) No



Assignments

- ✱ Module Week 4
- ✱ Next time: More random variable,
Expectations, Variance

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
You!*

